1. What do you mean by a geographic information system? Briefly explain the different components of a GIS. A Geographic Information System (GIS) is a computer-based tool that analyzes, stores, manipulates, and displays geographically referenced information. It acts as a Decision Support System that accepts large volumes of spatial data derived from a variety of sources and attaches this data to unique locations.

The five key components of a GIS are:

- **Hardware:** The computer systems on which the GIS operates. This ranges from centralized servers and workstation computers to desktop machines and even smart mobile devices.
- **Software:** Provides the functions and tools needed to store, analyze, and display geographic information. A good GIS software will have strong data input capabilities, database management systems, and tools to support geographic queries and visualizations.
- **Data:** This is possibly the most important component of GIS. It includes spatial data (which holds the exact geographic location of an object) and non-spatial/attribute data (tabular data describing what the object is).
- **People:** GIS technology requires people to manage the system and develop plans for applying it. This includes both GIS experts (who design, collect, and maintain the databases) and the end-users or decision-makers who utilize the final output.
- **Procedures (Methods):** A successful GIS operates according to well-designed plans, business rules, and models. This includes algorithms, data input procedures, and analytical methods used to execute specific tasks.

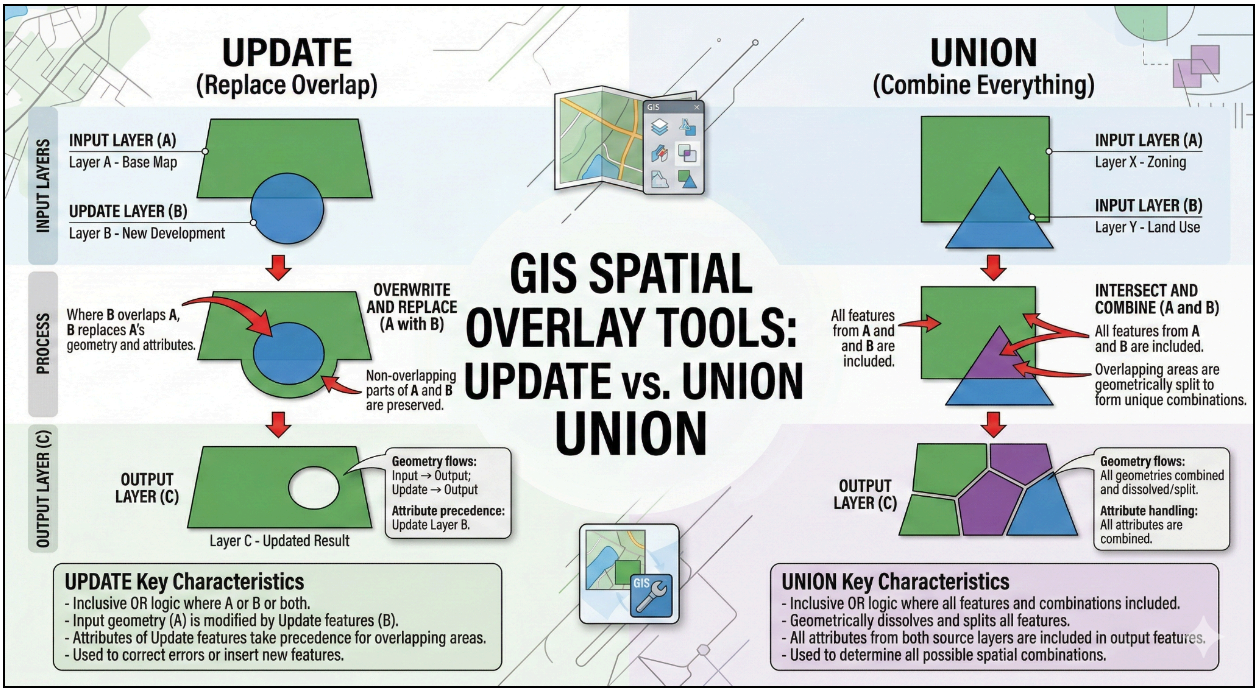
2. Explain the two data models used in GIS. The two primary data models used to represent spatial features in GIS are Raster and Vector:

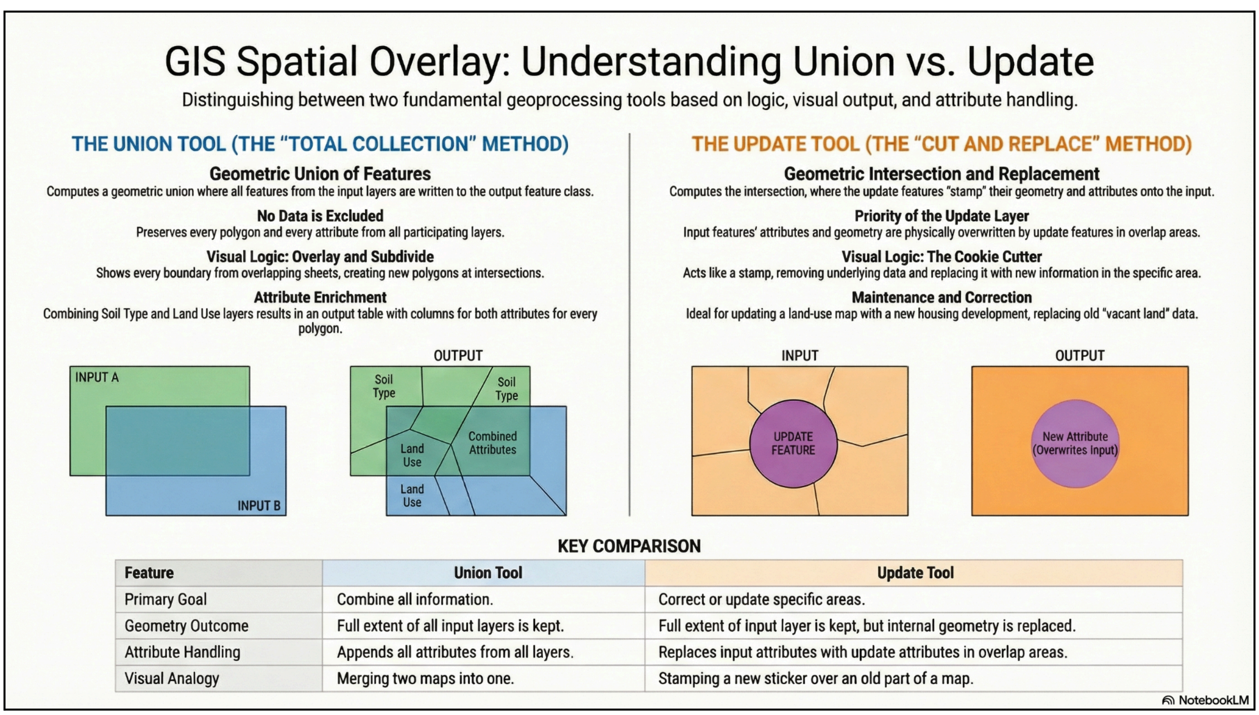
- **Raster Data Model:** This model represents the area of interest as a continuous surface divided into a regular grid of uniform cells (or pixels). Each cell has a single, associated value that represents the characteristics of the spatial phenomenon at that location (such as elevation in a Digital Elevation Model). The size of the cells determines the spatial resolution of the data.
- **Vector Data Model:** This model represents the world using discrete, primitive spatial data objects. These objects are defined by pairs of x,y coordinates in a spatial reference frame. The three primitive vector entities are **Points** (0-dimensional locations), **Lines/Polylines** (1-dimensional features like roads or rivers), and **Polygons** (2-dimensional areas like lakes or property boundaries). Unlike a raster cell, a single vector feature can have multiple attributes associated with it.

3. With the help of diagrams, distinguish between update and union tools of spatial overlay toolset used in GIS.

- **Union:** The Union tool computes a geometric union of the input features. All overlapping and non-overlapping features, along with their attributes from all input layers, are written to the output feature class.
- *Diagram Description:* Imagine two overlapping square polygons (an Input and another overlapping feature). The resulting Output diagram shows the outer boundary of both original squares merged together, with the internal intersecting boundary lines kept intact to form distinct new polygons where they overlapped.

- **Update:** The Update tool computes the geometric intersection of input features and update features. The attributes and geometry of the input features are completely updated/replaced by the update features in the output feature class where they overlap.
- *Diagram Description:* Imagine a rectangular "Input" feature and a circular "Update" feature placed over its bottom-center. The resulting Output diagram shows the original rectangle, but the bottom-center portion is completely replaced by the geometry of the circle, showing how the update feature "overwrites" the input area.





4. State the elements of a raster data model. The elements of a raster data model are:

- **Cell Value:** Each cell (pixel) contains a single value representing the characteristic of the spatial phenomenon at that location. Values can be categorical, 8-bit integers (0-255), floating points, or real values.
- **Cell Size (Spatial Resolution):** The dimension of the cell size representing the actual area covered on the ground. This determines the overall spatial resolution of the raster data.
- **Raster Bands:** To represent multiple values per cell (such as True Color or False Color images), the raster must be organized into multiple overlapping bands (e.g., Red, Green, and Blue bands).
- **Spatial Reference:** Each raster cell is identified by an ordered pair of coordinates (row and column numbers). This is generally tied to a projected coordinate system to georeference the raster in geographic space.